

Bayesian Econometrics

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Today

Bayesian Regression Analysis

Gibbs sampling

More flexible regression models

The Principle of Data Augmentation

Computational aspect: inefficiency factor

Metropolis-Hastings Algorithm

Bayesian Regression Analysis

Standard regression analysis (unknown homoskedastic variance)

- Data: observations $(\mathbf{x}_i, y_i)$, with predictor vector $\mathbf{x}_i = (x_{i1}, \dots, x_{id})$, $i = 1, \dots, N$.
- Model:

$$y_i = \mathbf{x}_i \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where the errors are iid Gaussian.

- $\boldsymbol{\beta} = (\beta_1, \dots, \beta_d)$ are the unknown regression coefficients, σ^2 is the unknown error variance.
- Inference: Posterior distribution $p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y})$ of $\boldsymbol{\beta}$ and σ^2 given observations $\mathbf{y}$:

$$p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}) \propto \underbrace{p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2)}_{\text{likelihood}} \underbrace{p(\boldsymbol{\beta}, \sigma^2)}_{\text{prior}}.$$

The likelihood of a regression model

- Rewrite the regression model using matrix notation and the multivariate normal distribution:

$$\mathbf{y} \sim \mathcal{N}_N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_N),$$

where $\mathbf{I}_N$ is the identity matrix, $\mathbf{y} = (y_1 \dots y_N)'$, $\mathbf{X} \in \mathbb{R}^{N \times d}$ is the design matrix, where the i th row contains the explanatory variables $\mathbf{x}_i$ corresponding to y_i .

- The likelihood $p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2)$ of $\mathbf{y}$ is equal to:

$$p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-N/2} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right).$$

Posterior distributions for a regression model

- The **likelihood considered as a function of σ^2** (holding β fixed) is proportional to the kernel of an **inverted Gamma distribution**:
 - $\Rightarrow$ the (conditionally) conjugate **prior** for σ^2 is an inverted Gamma distribution;
 - $\Rightarrow$ the (conditional) **posterior** $p(\sigma^2 \mid \beta, \mathbf{y})$ is an inverted Gamma distribution.
- The **log-likelihood is a multivariate square function in β** (holding σ^2 fixed), therefore the likelihood is the kernel of a **multivariate normal distribution** for β :
 - $\Rightarrow$ the (conditionally) conjugate **prior** for β is a multivariate normal distribution;
 - $\Rightarrow$ the conditional **posterior** $p(\beta \mid \sigma^2, \mathbf{y})$ is a multivariate normal distribution.

Inference: **Conjugate** linear regression analysis i

Closed-form posterior inference is possible if the prior covariance matrix of β depends on σ^2 :

- Likelihood:

$$p(\mathbf{y} \mid \beta, \sigma^2) = (2\pi\sigma^2)^{-N/2} \exp \left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)' (\mathbf{y} - \mathbf{X}\beta) \right)$$

- Priors: $\beta \mid \sigma^2 \sim \mathcal{N}_d(\mathbf{b}_0, \sigma^2 \mathbf{B}_0)$, $\sigma^2 \sim \mathcal{G}^{-1}(c_0, C_0)$.
- Posterior:

$$\begin{aligned} p(\beta, \sigma^2 \mid \mathbf{y}) &\propto p(\mathbf{y} \mid \beta, \sigma^2) p(\beta, \sigma^2) \\ &\propto (\sigma^2)^{-N/2} \exp \left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)' (\mathbf{y} - \mathbf{X}\beta) \right) \cdot \\ &\propto (\sigma^2)^{-d/2} \exp \left(-\frac{1}{2\sigma^2} (\beta - \mathbf{b}_0)' \mathbf{B}_0^{-1} (\beta - \mathbf{b}_0) \right) \cdot \\ &\propto (\sigma^2)^{-(c_0+1)} \exp \left(-\frac{C_0}{\sigma^2} \right) \end{aligned}$$

$$\Rightarrow p(\beta, \sigma^2 \mid \mathbf{y}) = p(\beta \mid \sigma^2, \mathbf{y}) p(\sigma^2 \mid \mathbf{y})$$

Conjugate linear Gaussian regression analysis ii

⇒ Then joint posterior density of (β, σ^2) is of closed form:

$$p(\beta, \sigma^2 | \mathbf{y}) = p(\beta | \sigma^2, \mathbf{y}) p(\sigma^2 | \mathbf{y})$$

where

$$\beta | \sigma^2, \mathbf{y} \sim \mathcal{N}_d(\mathbf{b}_N, \sigma^2 \mathbf{B}_N) \quad \text{and} \quad \sigma^2 | \mathbf{y} \sim \mathcal{G}^{-1}(c_N, C_N),$$

with parameters if $\mathbf{B}_0$ is large (flat prior) $\rightarrow$ the estimator converges to the frequentist estimate

$$\mathbf{B}_N^{-1} = \mathbf{B}_0^{-1} + \mathbf{X}'\mathbf{X}, \quad \mathbf{b}_N = \mathbf{B}_N(\mathbf{B}_0^{-1}\mathbf{b}_0 + \mathbf{X}'\mathbf{y})$$

$$c_N = c_0 + N/2, \quad C_N = C_0 + S_\varepsilon/2,$$

$$S_\varepsilon = (\mathbf{y} - \mathbf{X}\mathbf{b}_N)'(\mathbf{I}_N + \mathbf{X}\mathbf{B}_0^{-1}\mathbf{X}')^{-1}(\mathbf{y} - \mathbf{X}\mathbf{b}_N) = \mathbf{y}'\mathbf{y} + \mathbf{b}_0'\mathbf{B}_0^{-1}\mathbf{b}_0 - \mathbf{b}_N'\mathbf{B}_N^{-1}\mathbf{b}_N.$$

Non-conjugate linear Gaussian regression analysis

What if I want a different prior distribution?

- Assume that β and σ^2 are **independent** a priori:

$$\beta \sim \mathcal{N}_d(\mathbf{b}_0, \mathbf{B}_0), \quad \sigma^2 \sim \mathcal{G}^{-1}(c_0, C_0).$$

- $\Rightarrow$ the prior covariance $\mathbf{B}_0$ controls shrinkage toward $\mathbf{b}_0$;
- $\Rightarrow$ shrinkage of β towards irrelevant coefficients is achieved through $\mathbf{b}_0 = \mathbf{0}$ and a diagonal matrix $\mathbf{B}_0$ with small entries;
- $\Rightarrow$ shrinkage priors learn from the data the sizes of entries in $\mathbf{B}_0$.

Conditionally conjugate distributions

- No closed form joint posterior $p(\beta, \sigma^2 \mid \mathbf{y})$, but the conditional posteriors are of closed form:
 - Given σ^2 , the posterior $\beta \mid \sigma^2, \mathbf{y} \sim \mathcal{N}_d(\mathbf{b}_N, \mathbf{B}_N)$, where both $\mathbf{b}_N$ and $\mathbf{B}_N$ depend on σ^2 :

$$\mathbf{B}_N^{-1} = \mathbf{B}_0^{-1} + \mathbf{X}'\mathbf{X}/\sigma^2, \quad \mathbf{b}_N = \mathbf{B}_N(\mathbf{B}_0^{-1}\mathbf{b}_0 + \mathbf{X}'\mathbf{y}/\sigma^2),$$

- Given β , the posterior $\sigma^2 \mid \beta, \mathbf{y} \sim \mathcal{G}^{-1}(c_N, C_N)$, where C_N depends on β :

$$c_N = c_0 + N/2,$$

$$C_N = C_0 + 0.5\epsilon'\epsilon, \quad \epsilon = \mathbf{y} - \mathbf{X}\beta.$$

- A sampling based approach has to be applied!

Gibbs sampling

How to sample from this posterior?

- The posterior distribution $p(\theta \mid \mathbf{y})$ has no simple analytical form.
- It is not possible to efficiently produce **independent** random draws $\theta^{(1)}, \dots, \theta^{(M)}$ from $p(\theta \mid \mathbf{y})$.
- **But:** The posterior distribution can be characterized by **dependent** simulations from the posterior, which are obtained by using Markov chain Monte Carlo simulation methods.
- **Idea:** Important properties of the posterior are estimated from this sample as before by taking averages.

Markov chain Monte Carlo (MCMC) simulation

- A sample from the posterior $p(\theta \mid \mathbf{y})$ is obtained **indirectly by generating a realization of a first order Markov process** $\{\theta^{(m)}, m = 1, 2, \dots\}$, based on some starting value $\theta^{(0)}$.
- The **transition density** $k(\theta^{\text{new}} \mid \theta^{\text{old}})$ defines the conditional probability distribution of $\theta^{(m)}$ given $\theta^{(m-1)} = \theta^{\text{old}}$, i.e.

$$\theta^{(m)} \mid (\theta^{(m-1)} = \theta^{\text{old}}) \sim k(\theta^{\text{new}} \mid \theta^{\text{old}}),$$

and is selected in such a way, that the stationary distribution of $\theta^{(m)}$ is equal to the target distribution $p(\theta \mid \mathbf{y})$.

- The most widely used Markov chain Monte Carlo methods are:
 - **Gibbs sampling** [Geman and Geman, 1984, Casella and George, 1992]
 - **Metropolis-Hastings algorithm** [Hastings, 1970, Chib and Greenberg, 1995]
 - **Hamiltonian Monte Carlo** [Duante et al, 1987, Betancourt, 2018]

Gibbs sampling

- Gibbs sampling is a method to sample from a p -variate distribution

$$p(\mathbf{x}) = p(x_1, \dots, x_p)$$

- In each sweep of the sampler a single components x_j of $\mathbf{x}$ is updated by drawing from the full conditionals

$$x_j^{new} \sim p(x_j | x_{\setminus j}),$$

where each component of $x_{\setminus j} = (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_p)$ is fixed at the most recent draw. all other parameters are fixed in the current draw (1 unknown).

⇒ Gibbs sampling can be used to sample from the joint posterior $p(\boldsymbol{\theta} | \mathbf{y})$ of a multivariate parameter vector $\boldsymbol{\theta}$.

Posterior approximation by Gibbs sampling

Consider a model with multivariate parameter $\theta = (\theta_1, \dots, \theta_p)$.

- The posterior $p(\theta|\mathbf{y})$ is often **not of closed form**.
- **But**: often sampling from the **full conditional** distributions is feasible:

$$p(\theta_1|\theta_2, \dots, \theta_p, \mathbf{y})$$

$$p(\theta_2|\theta_1, \theta_3, \dots, \theta_p, \mathbf{y})$$

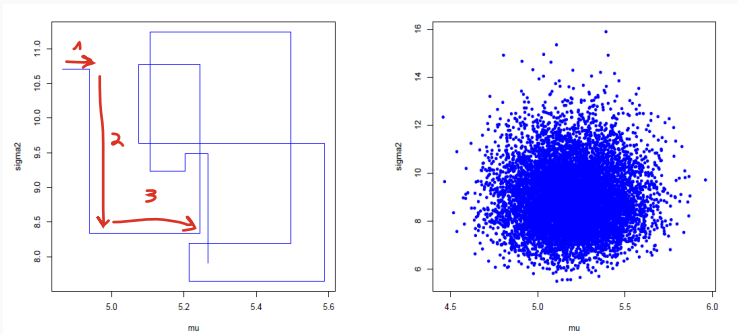
$$\vdots$$

$$p(\theta_p|\theta_1, \dots, \theta_{p-1}, \mathbf{y})$$

- Iterative sampling from the full conditionals constructs a **dependent** sequence of parameter values.
- The distribution of the sampled values converges to joint posterior $p(\theta|\mathbf{y})$.

Example: parameters of the Normal distribution

joint posterior density of μ and σ^2



100 simulated observations $y_i \sim \mathcal{N}(5, 9)$. Left: path of first 10 draws; right: 12000 draws of μ and σ^2

Gibbs sampling for a standard regression model

The **joint posterior density** $p(\beta, \sigma^2 \mid \mathbf{y})$ is approximated using a **two-block Gibbs sampler** where we sample iteratively from the full conditional distributions.

Gibbs sampling for regression model

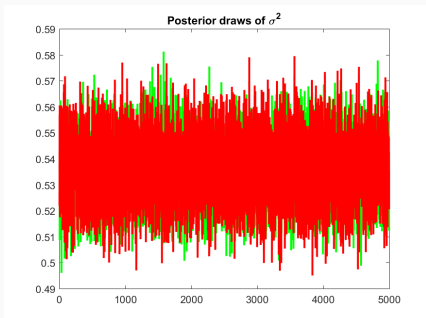
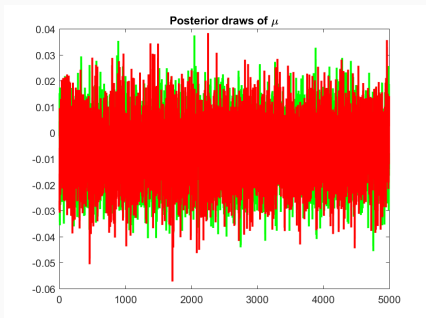
Choose a **starting value** $(\sigma^2)^{(0)}$ and repeat the following steps for $m = 1, \dots, M$:

- (a) **sample** $\beta^{(m)}$ from $p(\beta \mid (\sigma^2)^{(m-1)}, \mathbf{y})$, which is the multivariate normal distribution $\mathcal{N}_d(\mathbf{b}_N, \mathbf{B}_N)$ given on slide 8;
- (b) **sample** $(\sigma^2)^{(m)}$ from $p(\sigma^2 \mid \beta^{(m)}, \mathbf{y})$, which is the inverted Gamma distribution $\mathcal{G}^{-1}(c_N, C_N)$ given on slide 8.

Discard the first M_0 **draws as burn-in**. if we think the sampler needs some time to converge to the target (posterior) distribution

Application to exchange rate data

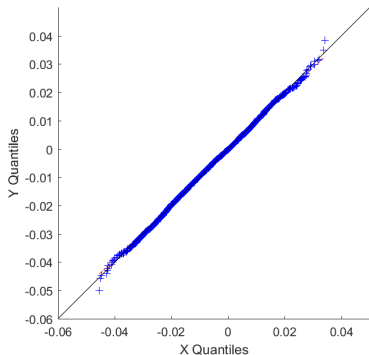
$M = 5000$ posterior draws of μ and σ^2 (including burn-in), based on priors $\mu \sim \mathcal{N}(0, 100)$ and $\sigma^2 \sim \mathcal{G}^{-1}(2.5, 1.5)$, where $E(\sigma^2) = 1$ is the prior expectation.



Verifying convergence: QQ plots

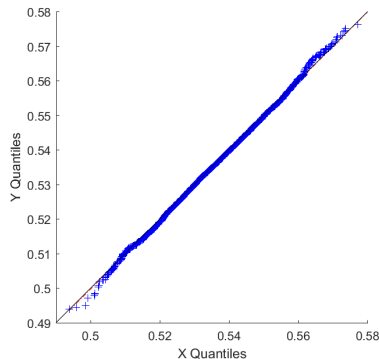
- Use a QQ plot to verify equality of posterior draws (excluding burn-in) from two independent runs with initial values $(\sigma^2)^{(0)} = 0.1$ and $(\sigma^2)^{(0)} = 10$.

posterior of μ



posterior of σ^2

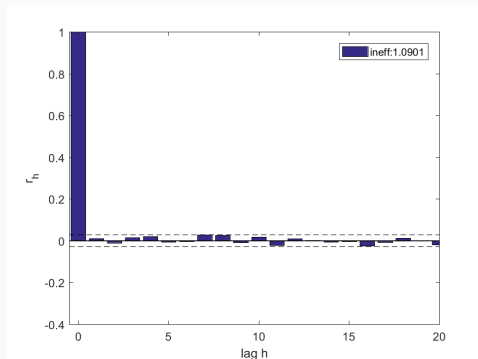
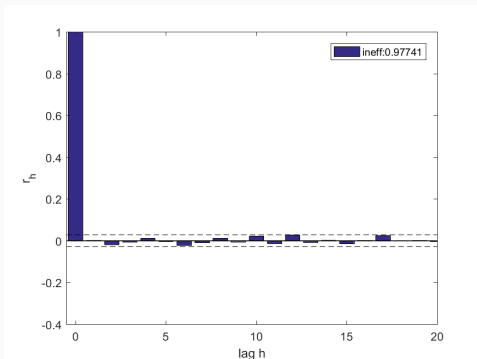
i.e. change the starting value



Posterior $p(\mu|\mathbf{y})$ (left hand side) and $p(\sigma^2|\mathbf{y})$ (right hand side)

Quantifying inefficiency

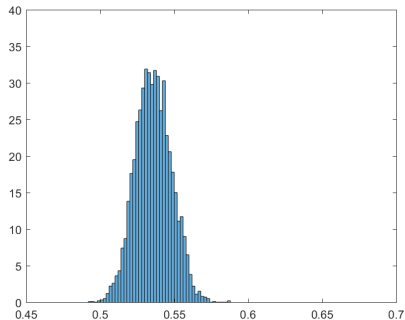
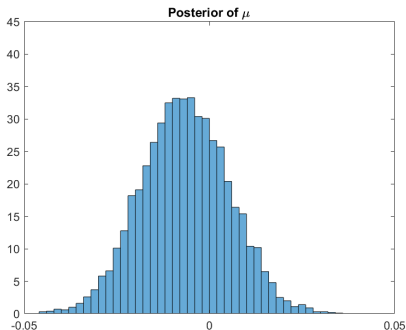
- ACF of posterior draws for μ (left hand side) and σ^2 (right hand side)



draws practically iid despite dependent sampling using a Markov chain!

Posterior distributions

- Estimate the marginal posteriors using a histogram of corresponding posterior draws



Posterior distributions of μ (left hand side) and σ^2 (right hand side)

Is the output reliable?

How large should M and M_0 be?

- Trace plot
- Auto-correlation function plot
- Independent chains and QQ-plot
- Effective sample size

Is the implementation correct?

- Test with a very large simulated data set: posterior should be close to the true value
- Test the entire sampler or parts of it using the “Getting it right” principle

With very skewed or heavy-tailed posteriors, even these may be difficult.

More flexible regression models

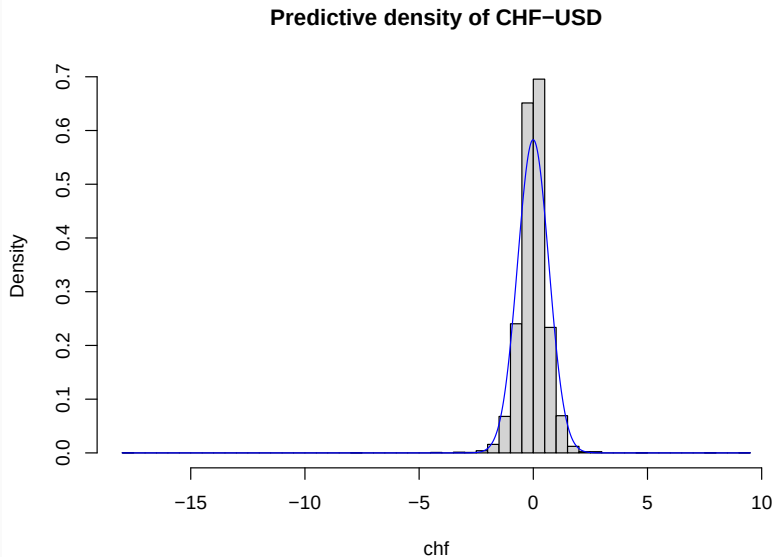
Flexible models in Bayesian econometrics

- Since the 1990s, **Markov chain Monte Carlo (MCMC)** became **the key ingredient** in the expansion of Bayesian econometrics in the entire Economics, Finance, and Business area [Geweke et al., 2011]:
- Freed Bayesian econometrics from the analytical restrictions inherent in conjugate analysis.
- Could handle models where the **likelihood is not available in closed form** (discrete choice models with random effects).
- Allowed **inference for non-linear functional** (impulse-response function) and **full prediction** (macro-economic and financial time series).

MCMC for flexible regression models

- **Regression analysis** with **Student- t errors** [Geweke, 1993]
- **Heteroscedastic errors**, e.g. SV modeling of errors in a time series context
- **Discrete-outcome models** for binary, multinomial, or count data, e.g. **probit** [Albert and Chib, 1993], **Tobit** [Chib, 1992]
- **Multinomial logit and probit models** [McCulloch and Rossi, 1994]
- **Time-varying parameter models**
[Carter and Kohn, 1994, Frühwirth-Schnatter, 1994]
- **Systems of equations**, e.g. **VAR-models** or **IV models**
- **Random effects models**

Fitting a normal distribution to exchange rate data



How well does the data fit the normal distribution?

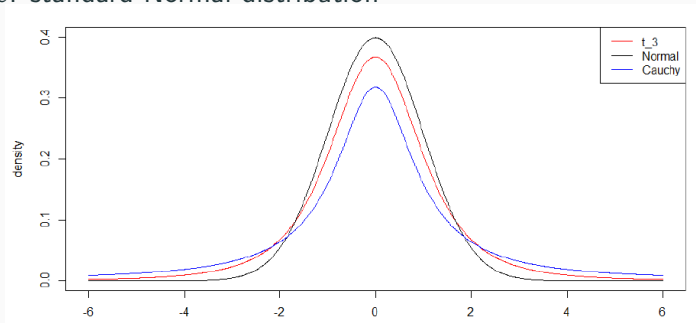
- Tail behavior of $\mathbf{y}$ not captured correctly by a normal distribution:

	α							
Q_α -percentiles	0.025	0.05	0.25	0.4	0.5	0.6	0.75	0.95
Normal	-1.3	-1.09	-0.45	-0.17	0	0.16	0.44	1.08
empirical	-1.18	-0.95	-0.32	-0.1	0.01	0.12	0.32	0.92
t_3	-1.22	-0.9	-0.3	-0.11	0	0.1	0.29	0.89

$\Rightarrow$ Model exchange rate data as $y_i \sim t_\nu(\mu, \sigma^2)$.

Student's t -distribution

- Standard t_ν - distribution is symmetric around zero.
- ν (the degrees of freedom) determine heaviness of the tails.
- Special cases:
 - $\nu = 1$: Cauchy distribution
 - $\nu \rightarrow \infty$: standard Normal distribution



- Applying the location-scale-transformation $X = \mu + \sigma T$, a three-parameter t -distribution μ, σ, ν results.

Modeling based on the Student- t Distribution

- Bayes' theorem yields the joint posterior distribution of $\theta = (\mu, \sigma^2, \nu)$ as:

$$p(\mu, \sigma^2, \nu \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mu, \sigma^2, \nu)p(\mu, \sigma^2, \nu),$$

where $p(\mathbf{y} \mid \mu, \sigma^2, \nu)$ is the likelihood of the data given μ , σ^2 and ν :

$$p(\mathbf{y} \mid \mu, \sigma^2, \nu) = \frac{\Gamma((\nu + 1)/2)^N}{\Gamma(\nu/2)^N \sqrt{\nu\pi}^N} \left(\frac{1}{\sigma^2} \right)^{N/2} \prod_{i=1}^N \left(1 + \frac{(y_i - \mu)^2}{\nu\sigma^2} \right)^{-(\nu+1)/2}.$$

- The likelihood regarded as a function of the unknown parameter $\theta = (\mu, \sigma^2, \nu)$ does not reduce to the kernel of a simple distribution and the posterior is not of closed form.

The Student- t as simple stochastic volatility model

Hierarchical representation of the Student- t distribution

The Student- t distribution $Y_i \sim t_\nu(\mu, \sigma^2)$ has a hierarchical representation with a **random rate** ω_i following a Gamma distribution:

$$\begin{aligned} Y_i \mid \mu, \sigma^2, \omega_i &\sim \mathcal{N}(\mu, \sigma^2 / \omega_i), \\ \omega_i \mid \nu &\sim \mathcal{G}(\nu/2, \nu/2). \end{aligned}$$

- The expectation of ω_i is equal to 1 and the variance of ω_i depends on the degrees of freedom:

$$\mathbb{E}(\omega_i \mid \nu) = 1, \quad \text{Var}(\omega_i \mid \nu) = \frac{2}{\nu}.$$

- As the degrees of freedom ν increases, ω_i converges to 1 with probability 1 and the marginal distribution of the data converges to the normal distribution $\mathcal{N}(\mu, \sigma^2)$.

The Principle of Data Augmentation

Latent variable representations

Three examples of latent variable models

- Economically meaningful but unobserved values
 - Business cycle
 - Time-varying variance/covariance
 - ...
- Artificial variables that simplify computation
 - Alternative representation for the Student- t distribution
 - Indicator that denotes whether covariate is zero (“insignificant”)
 - ...
- Missing data

Idea: Joint Bayesian inference for latent variables and parameters!

The Principle of Data Augmentation

- **Introduce latent auxiliary variables $\mathbf{z}$** such that the posterior $p(\boldsymbol{\theta} \mid \mathbf{y})$ is the marginal density of the joint posterior $p(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y})$:
$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \int p(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y}) d\mathbf{z}.$$
- **Add sampling of $\mathbf{z}$** as an additional block in Gibbs sampling
- Gibbs sampling yields **draws $(\boldsymbol{\theta}^{(m)}, \mathbf{z}^{(m)})$** from the joint posterior $p(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y})$.
- The (marginal) draws $\boldsymbol{\theta}^{(m)}$ obtained by **dropping $\mathbf{z}^{(m)}$** are draws from the marginal posterior $p(\boldsymbol{\theta} \mid \mathbf{y})$; no analytical integration necessary to obtain the marginal distribution.

The Principle of Data Augmentation

Gibbs sampling through data augmentation

Choose a starting value for $\theta^{(0)}$ and repeat the following steps for $m = 1, \dots, M$:

- (a) sample $z^{(m)}$ from $p(z \mid \theta^{(m-1)}, \mathbf{y})$,
- (b) sample $\theta^{(m)}$ from $p(\theta \mid z^{(m)}, \mathbf{y})$,

and remove M_0 draws as burn-in.

- Often the conditional density $p(\theta \mid z, \mathbf{y})$ given z is of closed form and therefore easy to sample from.
- The sampling order can be reversed and sampling started with an initial value $z^{(0)}$.

Student- t as simple stochastic volatility model

Hierarchical representation of the Student- t distribution

The Student- t distribution $Y_i \sim t_\nu(\mu, \sigma^2)$ has a hierarchical representation with a **random rate** ω_i following a Gamma distribution:

$$Y_i \mid \mu, \sigma^2, \omega_i \sim \mathcal{N}(\mu, \sigma^2 / \omega_i),$$
$$\omega_i \mid \nu \sim \mathcal{G}(\nu/2, \nu/2).$$

- The expectation of ω_i is equal to 1 and the variance of ω_i depends on the degrees of freedom:

$$\mathbb{E}(\omega_i \mid \nu) = 1, \quad \text{Var}(\omega_i \mid \nu) = \frac{2}{\nu}.$$

- As the degrees of freedom ν increases, ω_i converges to 1 in probability and the marginal distribution of the data converges to the normal distribution $\mathcal{N}(\mu, \sigma^2)$.

Bayesian Estimation for Student- t using data augmentation

- Data augmentation is based on introducing the sequence $\mathbf{z} = (\omega_1, \dots, \omega_N)$ as latent auxiliary variables.
- The posterior distribution is

$$p(\omega_1, \dots, \omega_N, \mu, \sigma^2 \mid \mathbf{y}),$$

and within a Gibbs sampler, the scaling factors $\mathbf{z} = (\omega_1, \dots, \omega_N)$ are sampled jointly with μ and σ^2 .

- For a **known** degrees of freedom parameter ν , full conditional Gibbs sampling is carried out in three blocks, where all conditional densities are of closed form.

Example: Regression models with Student- t errors

- A **regression model with Student- t errors**, i.e.

$$y_i = \mathbf{x}_i \boldsymbol{\beta} + \tilde{\varepsilon}_i, \quad \tilde{\varepsilon}_i \sim t_\nu(0, \sigma^2), \quad (1)$$

has the following hierarchical representation with latent variable ω_i :

$$\begin{aligned} y_i &= \mathbf{x}_i \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2 / \omega_i), \\ \omega_i &\sim \mathcal{G}(\nu/2, \nu/2). \end{aligned} \quad (2)$$

- Whereas model (1) is difficult to estimate due to the presence of the Student- t density, data augmentation makes estimation of model (2) pretty easy.
- Representation (2) also helps to understand the (marginal) model and points at possible modifications/extensions.

Gibbs sampling for a known degree of freedom ν

Gibbs sampling for a regression model with t_ν errors

Let $\mathbf{z} = (\omega_1, \dots, \omega_N)$ and σ^2 be some starting values (e.g. $\omega_i = 1$ for all i and σ^2 estimated from OLS residuals). Repeat the following steps and discard the burn-in:

- (a) Sample $\boldsymbol{\beta}$ from $\boldsymbol{\beta} \mid \sigma^2, \mathbf{z}, \mathbf{y} \sim \mathcal{N}_d(\mathbf{b}_N, \mathbf{B}_N)$ (with $\mathbf{b}_N, \mathbf{B}_N$ on slide 35);
- (b) sample σ^2 from $\sigma^2 \mid \boldsymbol{\beta}, \mathbf{z}, \mathbf{y} \sim \mathcal{G}^{-1}(c_N, C_N)$ (with c_N, C_N on slide 36);
- (c) **NEW:** Sample $\omega_i \mid \boldsymbol{\beta}, \sigma^2, y_i$ independently for each $i = 1, \dots, N$ from

$$\omega_i \mid \boldsymbol{\beta}, \sigma^2, y_i \sim \mathcal{G}(\nu/2 + 1/2, \nu/2 + r_i^2/2),$$

where r_i are the standardized residuals $r_i = (y_i - \mathbf{x}_i \boldsymbol{\beta}) / \sqrt{\sigma^2}$.

Regression models with heteroskedastic errors

- Consider a regression model with heteroskedastic errors with $\sigma_1^2, \dots, \sigma_N^2$ known (e.g. for Student- t , $\sigma_i^2 = \sigma^2/\omega_i$, but more general models possible):

$$y_i = \mathbf{x}_i \beta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2).$$

- This model can be transformed into a “pseudo” regression model with unit error variance:

$$\begin{aligned} y_i^* &= \mathbf{x}_i^* \beta + \varepsilon_i^*, \quad \varepsilon_i^* \sim \mathcal{N}(0, 1), \\ y_i^* &= \frac{y_i}{\sqrt{\sigma_i^2}}, \quad \mathbf{x}_i^* = \frac{1}{\sqrt{\sigma_i^2}} \mathbf{x}_i. \end{aligned} \tag{3}$$

- Works also for more general volatility models for σ_i^2 such as stochastic volatility.**

Regression models with heteroskedastic errors 2

- Perform Bayesian inference with respect to β in the “pseudo” regression model (3):
- under the flat prior $p(\beta) \propto c$:

$$\beta \mid \sigma^2, \mathbf{y} \sim \mathcal{N}_d(\mathbf{b}_N, \mathbf{B}_N),$$

$$\mathbf{b}_N = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{W} \mathbf{y}), \quad \mathbf{B}_N = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1},$$

if $\mathbf{W}=\mathbf{I}$ we have the OLS estimator

$$\mathbf{W} = \text{Diag}(1/\sigma_1^2, \dots, 1/\sigma_N^2).$$

- **Relation to WLS (weighted least squares)** with weight matrix $\mathbf{W}$ being learned from the data on the fly.

Regression models with heteroskedastic errors 3

- For informative priors $\beta \sim \mathcal{N}_d(\mathbf{b}_0, \mathbf{B}_0)$, a similar form is obtained:

$$\mathbf{b}_N = \mathbf{B}_N(\mathbf{X}^\top \mathbf{W} \mathbf{y} + \mathbf{B}_0^{-1} \mathbf{b}_0), \quad \mathbf{B}_N = (\mathbf{X}^\top \mathbf{W} \mathbf{X} + \mathbf{B}_0^{-1})^{-1}.$$

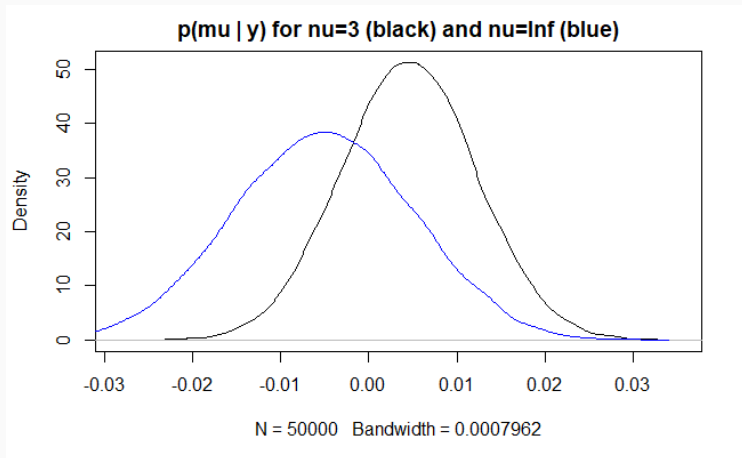
- For prior $\sigma^2 \sim \mathcal{G}^{-1}(c_0, C_0)$, the posterior $\sigma^2 \mid \beta, \omega, \mathbf{y} \sim \mathcal{G}^{-1}(c_N, C_N)$, where

$$c_N = c_0 + N/2,$$

$$C_N = C_0 + 0.5 \boldsymbol{\varepsilon}' \boldsymbol{\varepsilon}, \quad \varepsilon_i = \sqrt{\omega_i}(y_i - \mathbf{X}_{i,\cdot} \beta)$$

Application to exchange rate data

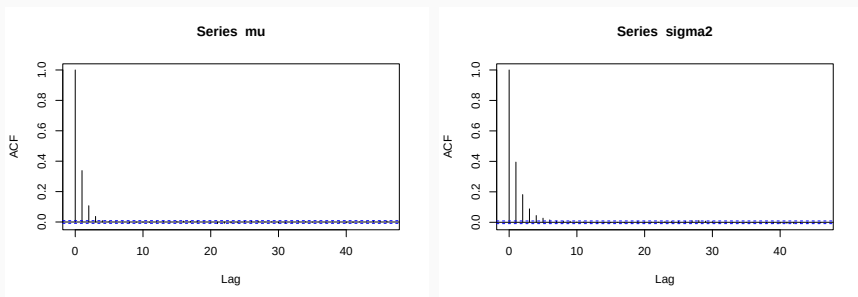
Posterior draws of μ , based on $Y_i \sim t_\nu(\mu, \sigma^2)$ with $\nu = 3$ and $\nu = \infty$.



**Computational aspect:
inefficiency factor**

Quantifying inefficiency

ACF of posterior draws for μ (left) and σ^2 (right)



ACF of posterior draws for μ and σ^2 show serial dependence

Classical versus Markov chain Monte Carlo Methods

Consider an estimator for $E(h(\theta) \mid \mathbf{y})$ derived from M draws $\theta^{(1)}, \dots, \theta^{(M)}$ from the posterior density $p(\theta \mid \mathbf{y})$:

$$E(h(\theta) \mid \mathbf{y}) \approx \overline{h(\theta)} = \frac{1}{M} \sum_{m=1}^M h(\theta^{(m)}).$$

Variance of $\overline{h(\theta)}$:

- Classical Monte Carlo integration uses a sample of **independent draws** and the variance of the estimator is given by:

$$\text{Var}(\overline{h(\theta)}) = \frac{1}{M} \text{Var}(h(\theta) \mid \mathbf{y}),$$

- Markov chain Monte Carlo (**MCMC**) methods use a sample that is the realization of an irreducible, aperiodic Markov chain. The **dependence among the draws** influences the variance of the estimator:

$$\text{Var}(\overline{h(\theta)}) = \frac{1}{M} \Omega_0(h(\theta) \mid \mathbf{y}).$$

$\Omega_0 > \text{Var}$

Inefficiency factors

- $\Omega_0(h(\theta) | \mathbf{y})$ is the spectral density of the process $h(\theta^{(m)})$ at 0 and differs from $\text{Var}(h(\theta) | \mathbf{y})$ by the following factor [Geweke, 1992]:

$$IF = \frac{\Omega_0(h(\theta) | \mathbf{y})}{\text{Var}(h(\theta) | \mathbf{y})} = 1 + 2 \sum_{s=1}^{\infty} \rho_s,$$

e.g. $M=1000$, $IF=1.2$
 $\Rightarrow$ 1200 dependent draws
needed to get the same
information as 1000 indep.
draws

where ρ_s is the autocorrelation function of the process $h(\theta^{(m)})$ at lag s .

- This ratio is called the **inefficiency factor** as it answers the following question:
 - By which factor has the number of draws from the Markov chain be multiplied to have a Monte Carlo error comparable to M iid draws?
 - A factor bigger than 1 indicates that MCMC sampling is less efficient than iid sampling.
- In our CHF application with $\nu = 3$, σ^2 has the worst inefficiency factor, about 2.8.

Inefficiency factor and effective sample size

Practical implementation:

- First, convergence of the chain has to be verified
 - fast method is a trace plot; it should look like a stationary process
 - thorough method is via independent chains and QQ-plots
- Then, for a posterior sample stored in vector v , the inefficiency factor can be computed in R as
$$\text{length}(v) / \text{coda::effectiveSize}(v)$$

The effective sample size ESS is a related concept.

- If ESS is k , then intuitively our posterior sample of size n contains as much information about the posterior distribution as k iid draws from the posterior distribution.
- Effective sample size multiplied by the inefficiency factor is (by definition) the sample size.

Estimate ν

Now we would like to sample $\nu \dots$

- $\implies$ But we can't find a conjugate distribution! we cannot find the closed-form of the cond. dist. of ν
- $\implies$ We do not have a Gibbs sampling step for ν !

Metropolis-Hastings Algorithm

The Metropolis-Hastings algorithm is perhaps the most popular example of a Markov chain Monte Carlo (MCMC) method in statistics.

Rfc`\_qga nrm`jc k rf_r gr qmjt cq gq rm nrm t g b c _ k c r f m b d m p q _ k n j g l e d p m k _ l
p`grp\_pw nrm``g j g r w b g q r p g` s r g m l,

The idea is to define a Markov chain, in such a way that the stationary distribution of the Markov chain is in fact .

Metropolis-Hastings Algorithm

For a general parameter θ :

- We would like to construct a Markov chain $\{\theta^{(m)}, m = 1, \dots, M\}$ with stationary distribution $p(\theta \mid \mathbf{y})$.
- The **Metropolis-Hastings algorithm** creates a transition from $\theta^{(m-1)}$ to $\theta^{(m)}$ as follows:
 - An (arbitrary) **proposal density** $q(\theta^{\text{new}} \mid \theta^{\text{old}})$ is selected which is used for proposing a value θ^{new} for $\theta^{(m)}$, conditional on $\theta^{(m-1)} = \theta^{\text{old}}$:

$$\theta^{(m)} \mid \theta^{(m-1)} = \theta^{\text{old}} \sim q(\theta^{\text{new}} \mid \theta^{\text{old}}).$$

- The proposed value θ^{new} is **accepted only with a certain probability** $\alpha(\theta^{\text{new}} \mid \theta^{\text{old}})$.
- If the proposed value is accepted, then $\theta^{(m)} = \theta^{\text{new}}$, otherwise the chain does not move: $\theta^{(m)} = \theta^{\text{old}} = \theta^{(m-1)}$.

Metropolis-Hastings Algorithm

- Metropolis-Hastings theorem: the following **acceptance probability**

$$\alpha(\theta^{\text{new}} \parallel \theta^{\text{old}}) = \min \left(1, \frac{p(\theta^{\text{new}} \parallel \mathbf{y})q(\theta^{\text{old}} \parallel \theta^{\text{new}})}{p(\theta^{\text{old}} \parallel \mathbf{y})q(\theta^{\text{new}} \parallel \theta^{\text{old}})} \right),$$

guarantees that $p(\theta^{\text{new}} \parallel \mathbf{y})$ is the invariant distribution of this Markov chain whenever the transition density $q(\theta^{\text{new}} \parallel \theta^{\text{old}})$ is aperiodic and irreducible.

- The computation of $\alpha(\theta^{\text{new}} \parallel \theta^{\text{old}})$ is possible from the right hand side of Bayes' theorem:

$$\alpha(\theta^{\text{new}} \parallel \theta^{\text{old}}) = \min \left(1, \frac{p(\mathbf{y} \parallel \theta^{\text{new}})p(\theta^{\text{new}})q(\theta^{\text{old}} \parallel \theta^{\text{new}})}{p(\mathbf{y} \parallel \theta^{\text{old}})p(\theta^{\text{old}})q(\theta^{\text{new}} \parallel \theta^{\text{old}})} \right),$$

because **the normalizing constant of the posterior density cancels**.

$p(\mathbf{y})$ cancels (it is in the denominator of both conditional densities - old and new)

Practical implementation

Metropolis-Hastings step

- Given $\theta^{\text{old}} = \theta^{(m-1)}$, propose a value $\theta^{\text{new}} \mid \theta^{\text{old}} \sim q(\theta^{\text{new}} \mid \theta^{\text{old}})$.
- Draw U from the uniform distribution $\mathcal{U}[0, 1]$ and accept θ^{new} , if

$$U \leq \frac{p(\mathbf{y} \mid \theta^{\text{new}})p(\theta^{\text{new}})q(\theta^{\text{old}} \mid \theta^{\text{new}})}{p(\mathbf{y} \mid \theta^{\text{old}})p(\theta^{\text{old}})q(\theta^{\text{new}} \mid \theta^{\text{old}})},$$

where $p(\mathbf{y} \mid \theta)$ is the likelihood function and $p(\theta)$ is the prior (for numerical reasons, we often take the log of both sides and work with that).

- If θ^{new} is accepted, then $\theta^{(m)} = \theta^{\text{new}}$;
- otherwise the sampler remains in the old state, i.e. $\theta^{(m)} = \theta^{(m-1)}$.

Requirements for MH

- Select a proposal distribution q , which may depend on the current values of θ .
- We need to be able to sample from q .
- We need to be able to evaluate the density q .
- We need to be able to evaluate the unnormalized posterior density.

Random Walk MH

- Theoretically, (almost) any proposal density $q(\theta^{\text{new}} \mid \theta^{\text{old}})$ delivers samples from the target distribution [Tierney, 1994]. However, the efficiency of the method may depend crucially on choosing an appropriate proposal.
- Commonly a **random walk proposal** is used for proposing θ^{new} :

$$\theta^{\text{new}} \mid \theta^{\text{old}} \sim \mathcal{N}(\theta^{\text{old}}, C),$$

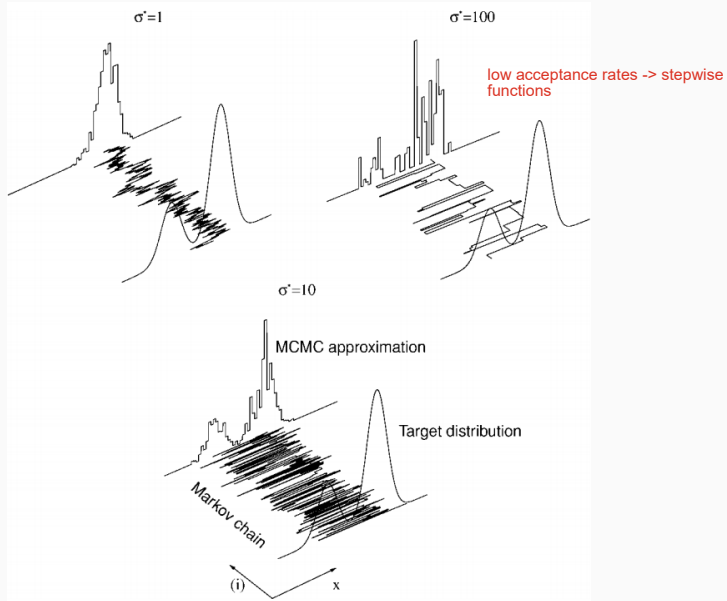
with fixed scale matrix C .

- As the corresponding proposal density is symmetric, $q(\theta^{\text{new}} \mid \theta^{\text{old}}) = q(\theta^{\text{old}} \mid \theta^{\text{new}})$, the acceptance ratio **simplifies** to the ratio of the non-normalized posterior $p^*(\theta \mid \mathbf{y}) = p(\mathbf{y} \mid \theta)p(\theta)$ evaluated at the new and the old value:

$$\alpha(\theta^{\text{new}} \mid \theta^{\text{old}}) = \min \left(1, \frac{p(\mathbf{y} \mid \theta^{\text{new}})p(\theta^{\text{new}})}{p(\mathbf{y} \mid \theta^{\text{old}})p(\theta^{\text{old}})} \right).$$

Tuning the random walk proposal

- **Moving in the right direction:**
 - A parameter θ^{new} with higher posterior density than θ^{old} is accepted in any case.
 - θ^{new} with lower posterior density than θ^{old} is accepted with probability proportional to the posterior density ratio.
- The **scale C** of a random walk proposal density has to be chosen with care:
 - A **small** scale C generates small moves $\theta^{\text{new}} - \theta^{\text{old}}$ with high acceptance rates, but also **high inefficiency factors**.
 - The limiting case where C tends to 0 has an acceptance rate of 1, but the inefficiency factor is infinite as the resulting Markov chain does not move.
 - A **large** scale C generates large moves $\theta^{\text{new}} - \theta^{\text{old}}$ and often propose a value far out in the tails of the distribution, giving generally **small acceptance rates**.
 - In practice, the scale of the proposal is tuned until the acceptance rate is in the order of **0.2-0.4** [Roberts and Rosenthal, 1998]



Student- t errors with unknown degrees of freedom

MH within Gibbs

(*) Sample β , σ^2 , and $\omega_1, \dots, \omega_N$ in some order; then

(d) Sample ν from $p(\nu \mid \omega_1, \dots, \omega_N)$ using an MH-step with a random walk proposal for $\log \nu^{\text{new}}$: we use lognormal for ν to avoid negative values

(d-1) Propose ν^{new} through $\log \nu^{\text{new}} \mid \nu^{\text{old}} \sim \mathcal{N}(\log \nu^{\text{old}}, c_\nu^2)$, where $\nu^{\text{old}} = \nu^{(m-1)}$.

(d-2) Accept $\nu^{(m)} = \nu^{\text{new}}$ with probability

$$\alpha(\nu^{\text{new}} \mid \nu^{\text{old}}) = \min \left(1, \frac{p(\omega \mid \nu^{\text{new}})p(\nu^{\text{new}})\nu^{\text{new}}}{p(\omega \mid \nu^{\text{old}})p(\nu^{\text{old}})\nu^{\text{old}}} \right),$$

where $p(\omega \mid \nu)$ is the Gamma “likelihood” function for $\omega = (\omega_1, \dots, \omega_N)$ (it is a likelihood from ν ’s perspective) and $p(\nu)$ is the prior.

(d-3) Otherwise, $\nu^{(m)} = \nu^{(m-1)}$.

Student- t errors with unknown degrees of freedom

- Since the proposal density $q(\nu^{\text{new}} \mid \nu^{\text{old}})$ is the density of the log-normal distribution with mean $\log \nu^{\text{old}}$, the **ratio of the two proposal densities**,

$$\frac{q(\nu^{\text{old}} \parallel \nu^{\text{new}})}{q(\nu^{\text{new}} \parallel \nu^{\text{old}})} = \frac{\nu^{\text{new}}}{\nu^{\text{old}}},$$

is not equal to 1.

- Implementation is straightforward, but **tuning** is necessary to choose the scale c_ν of the proposal densities.

Application to exchange rate data

- Model exchange rate data as

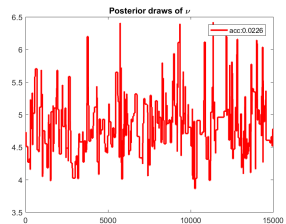
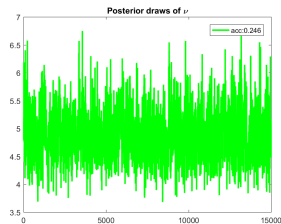
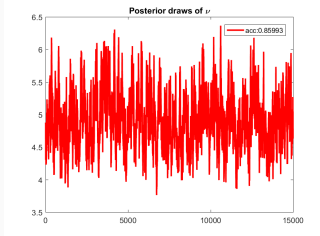
$$Y_i \sim t_\nu(\mu, \sigma^2)$$

with (μ, σ^2, ν) unknown.

- Priors:
 - $\mu \sim \mathcal{N}(0, 100)$;
 - $\sigma^2 \sim \mathcal{G}^{-1}(c_0, (c_0 - 1))$, where $c_0 = 2.5$.
 - $\nu \sim \mathcal{E}(1)$.
- Use random walk Metropolis-Hastings steps for ν and tune the proposal scale c_ν .
- $M = 15000$ draws (burn-in 2000).

Application to exchange rate data

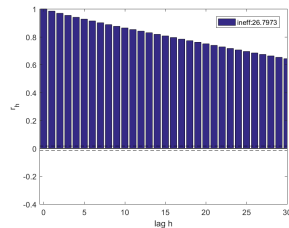
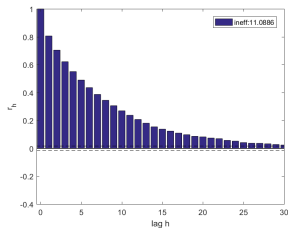
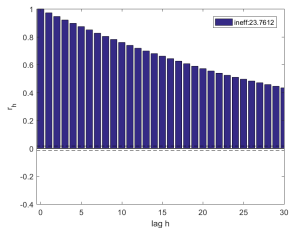
Posterior draws for degrees of freedom ν



$c_\nu = 0.05$ (left-hand side), $c_\nu = 0.5$ (middle), $c_\nu = 5$ (right-hand side)

Quantifying inefficiency

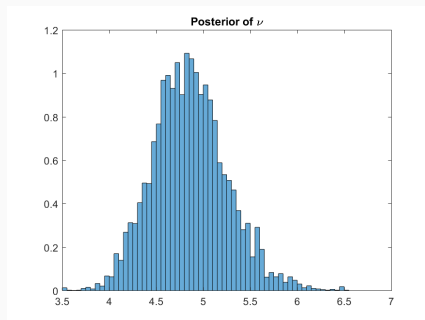
ACF of posterior draws for degrees of freedom ν



for extreme acceptance rates the following obs. is often the same as the previous one -> high ACF

$c_\nu = 0.05$ (left-hand side), $c_\nu = 0.5$ (middle), $c_\nu = 5$ (right-hand side)

Learning the degrees of freedom ($c_\nu = 0.5$)



Posterior $p(\nu|\mathbf{y})$, $E(\nu|\mathbf{y}) = 4.85$, 95%-HPD region: $[4.14, 5.72]$

“Getting it right” software test

Based on results by [Geweke, 2004], if there is an MCMC algorithm $\mathcal{A}$ that produces a sample from the posterior distribution $p(\theta \mid y)$ and if it is possible to sample y from the likelihood, then the following algorithm results in a sample $\theta \sim p(\theta)$.

Geweke test

Initialize $\theta^{(0)}$ and repeat the following step for $m = 1, \dots, M$:

1. Sample new data $y^{(m)} \parallel \theta^{(m-1)}$ from the likelihood $p(y \mid \theta = \theta^{(m-1)})$;
2. Sample new parameter values $\theta^{(m)} \parallel y^{(m)}$ using algorithm $\mathcal{A}$;

Discard the first M_0 values;

That is, if the data is regenerated in every round, then the resulting “posterior” for θ is the prior for θ . This can be used to test the correctness of your implementation.

since there is no other information, we should get the prior eventually

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